



CONCEPT



Rounding rules

We round down if the digit is 4 or less. We round up if the digit is 5 or more.

When both numbers are rounded down
the answer is an underestimate.

$$319 + 443 = 762$$

$$300 + 400 = 700$$

The estimate is less than the actual answer.

When both numbers are rounded up
the answer is an overestimate.

$$482 + 257 = 739$$

$$500 + 300 = 800$$

The estimate is greater than the actual answer.

1 a

	H	T	U
		7	5
+	6	5	4

b

	H	T	U
		8	6
+	5	4	7

c

	H	T	U
		6	6
+		7	5

d

		1	2	6
+	5	7	8	

e

		3	4	9
+	2	5	7	

f

		5	7	9
+	2	1	6	

g

		3	4	6
+	3	7	7	

h

		8	3	5
+	1	4	8	

i

		1	5	8
+	6	7	3	

j

		3	2	9
+	2	6	6	

k

		1	0	7
+	6	9	3	

l

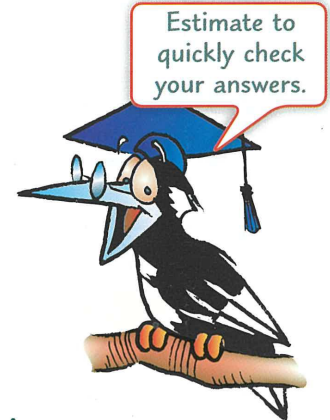
		4	4	5
+	1	9	8	

m

		5	9	8
+	2	5	3	

n

		2	8	9
+	3	7	9	



2 Use the strategy shown on the right to do these.

a $197 + 686$

+3	-3
----	----

b $445 + 198$

-2	+2
----	----

c $407 + 369$

--	--

d $295 + 288$

--	--

e $465 + 283$

+3-20	-3+20
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f $378 + 188$

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Explain the strategy used here.

$$316 + 147 = (316 - 3) + (147 + 3)$$

$$= 313 + 150$$

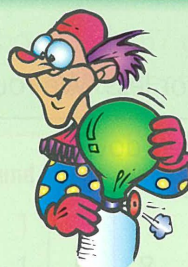
$$= 463$$



CONCEPT



583 apples and 179 oranges were sold.
How many pieces of fruit were sold altogether?



All of these are
ways of setting
out the same sum.

H	T	U
1	1	
5	8	3
+	1	7
7	6	2

1	1	
5	8	3
+	1	7
7	6	2

(or) $500 + 80 + 3$
 $100 + 70 + 9$
 $\begin{array}{r} 1 \ 2 \\ + 6 \ 0 \ 0 \\ \hline 7 \ 6 \ 2 \end{array}$

(or) $583 + 179$
 $= 583 + 179$
 $\quad (-1) \quad (+1)$
 $= 582 + 180$
 $\quad (-20) \quad (+20)$
 $= 562 + 200$
 $= 762$

762 pieces of fruit were sold altogether.

1 a $\begin{array}{r} 4 \ 3 \ 6 \\ + 1 \ 5 \ 7 \\ \hline \end{array}$

b $\begin{array}{r} 4 \ 7 \ 6 \\ + 2 \ 4 \ 7 \\ \hline \end{array}$

c $\begin{array}{r} 2 \ 6 \ 4 \\ + 2 \ 4 \ 6 \\ \hline \end{array}$

d $\begin{array}{r} 4 \ 5 \ 9 \\ + 3 \ 5 \ 6 \\ \hline \end{array}$

e $\begin{array}{r} 8 \ 8 \ 8 \\ + 1 \ 0 \ 7 \\ \hline \end{array}$

f $\begin{array}{r} 3 \ 0 \ 9 \\ + 1 \ 9 \ 3 \\ \hline \end{array}$

g $\begin{array}{r} 4 \ 4 \ 4 \\ + 4 \ 5 \ 6 \\ \hline \end{array}$

h $\begin{array}{r} 4 \ 7 \ 0 \\ + 2 \ 6 \ 0 \\ \hline \end{array}$

i $\begin{array}{r} 5 \ 7 \ 8 \\ + 1 \ 4 \ 4 \\ \hline \end{array}$

j $\begin{array}{r} 7 \ 3 \ 6 \\ + 1 \ 9 \ 9 \\ \hline \end{array}$

k $\begin{array}{r} 3 \ 8 \ 8 \\ + 3 \ 1 \ 7 \\ \hline \end{array}$

l $\begin{array}{r} 1 \ 9 \ 4 \\ + 2 \ 5 \ 5 \\ \hline \end{array}$

2 Estimate each answer, then use the estimate to check your answer to the problem.

a There were 375 men and 486 women employed in the company.
How many people were employed in the company altogether?

people

b We packed 563 boxes of toys on Thursday and 187 on Friday.
How many boxes of toys did we pack altogether?

boxes

c Last year I collected 267 swap cards. This year I collected 100 less than that. How many did I collect in the two years?

cards

d I started a part-time job last week. I earned \$274 last week and \$198 this week. How much have I earned so far?

dollars

e I need to travel to my grandparents' home and back today.
They live 428 km away. How far must I travel?

km



We write 234 in a more useful way. $234 = 220 + 14$



CONCEPT

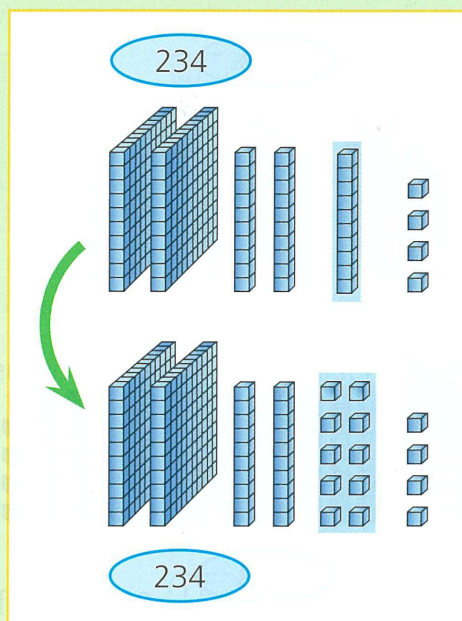


John had 234 stamps but 18 were damaged. How many were left?

$$\begin{array}{r} 234 \\ - 18 \\ \hline 216 \end{array}$$

one ten

We can't take 8 from 4, so we trade 1 ten for 10 ones.



$$\begin{array}{r} 20 \quad 14 \\ 200 + 30 + 4 \\ \text{take away} \\ 10 + 8 \end{array}$$

216 stamps were left.



- 1 a

	H	T	U
		8	0
-		4	7
- b

	H	T	U
		7	1
-		3	7
- c

	H	T	U
		1	5
-		2	5
- d

	H	T	U
		2	4
-		3	5
- e

	H	T	U
		5	5
-		2	3
- f

	H	T	U
		6	7
-		1	3
- g

	H	T	U
		5	7
-		3	2
- h

	H	T	U
		3	6
-		2	5
- i

	H	T	U
		4	7
-		1	4
- j

	H	T	U
		7	7
-		1	1
- k

	H	T	U
		5	6
-		2	4
- l

	H	T	U
		4	9
-		1	6
- m

	H	T	U
		8	7
-		4	3
- n

	H	T	U
		7	9
-		3	3
- o

	H	T	U
		6	5
-		3	0
- p

	H	T	U
		4	4
-		3	0

2 Check your answers to Question 1, i to p, by rounding each number to the nearest 100.

(i)

	H	T	U
		5	0
-		1	0

(j)

	H	T	U
		8	0
-		1	0

(k)

	H	T	U
		6	0
-		2	0

(l)

	H	T	U
-			

(m)

	H	T	U
-			

(n)

	H	T	U
-			

(o)

	H	T	U
-			

(p)

	H	T	U
-			